Chipalooza Challenge #4 Proposal: Buffered 6-Bit Resistor-String DAC

Team name: Flatline · **ICARUS team number:** 4

Attachments sent separately: designer CVs.

1. Type of IP block

A voltage-output, 6-bit resistor-string DAC with an integrated unity-gain output buffer, in Sky130. It is intended for low-frequency set points inside an SoC: programmable comparator thresholds, trim and calibration voltages, and bias set points.

2. Functional description

- **Resistor string:** 64 matched unit resistors between the external references VREFH and VREFL. The resistor string is inherently monotonic.
- **Tap selector:** a 6-level binary tree of CMOS transmission gates, with one level per data bit, so no decoder is needed. The selected tap is
$$VTAP = VREFL + D \times \frac{VREFH - VREFL}{64},$$
for D = 0 to 63.
- **Level shifters:** seven cross-coupled level shifters convert D5–D0 and EN from 1.8 V logic into complementary 0 / 3.3 V switch drives.
- **Output buffer:** a Sky130 transistor-level op-amp in unity-gain feedback. It isolates the high-impedance string (up to about 16 kΩ at mid-code) from the load.
- **Enable:** with EN high, the selected tap drives the buffer. With EN low, the buffer input switches to VREFL, giving a defined low output. The string and buffer remain powered.

Codes may be changed while EN is high. Brief glitches at multi-bit transitions are expected.

3. I/O list and harness resources

Pin	Type	Harness resource	Notes
D5–D0	Digital in, 1.8 V	6 control inputs	D5 = MSB
EN	Digital in, 1.8 V	1 control input	May map to the per-block harness enable

Pin	Type	Harness resource	Notes
VREFH	Analog in, 3.0V	Shared analog line	Sources the string current, about 42 µA
VREFL	Analog in, 0.3V	Shared analog line	Must sink the string current
VTAP	Analog test out	Shared analog line	Unbuffered tap voltage; test use only, needs a ≥10 MΩ meter
VOUT	Analog out	1 dedicated pad (requested)	Buffered output; must not be shared
IBIAS	Bias in	1 mirrored current, 1 µA	Bias for the buffer
VDDA / VDDD / VSS	Power	Per-slot switch	3.3 V analog, 1.8 V digital

Total: 7 of 24 control inputs, 0 status outputs, 3 of 4 shared analog lines, 1 dedicated pad, and 1 bias current. The voltage drop across a shared-line switch at 42 µA is about 0.2 mV (about 0.005 LSB), so placing the references on shared lines costs essentially no accuracy. VTAP lets us measure string linearity and buffer offset (VOUT − VTAP) separately. It is also a usable unbuffered fallback output.

4. Target specification

Conditions: VREFL = 0.300 V, VREFH = 3.000 V, load 100 kΩ // 20 pF (including pad, package, and fixture), over the operating range below unless noted. 1 LSB = 42.1875 mV. Errors are referred to the measured reference voltages.

Parameter	Min	Typ	Max	Unit
VDDA	3.135	3.3	3.465	V
VDDD	1.71	1.8	1.89	V
Temperature	−20	25	110	°C
IBIAS	0.8	1.0	1.2	µA

Parameter	Min	Typ	Max	Unit
Resolution		6		bit
Output range (code 0 to 63)		0.300 to 2.958		V
Absolute error, every code	−0.5		+0.5	LSB
DNL	−0.5		+0.5	LSB
INL (endpoint fit)	−0.5		+0.5	LSB
Monotonicity	Guaranteed			
Settling after a code change, to ±0.5 LSB		10	20	µs
Reference current		42.2	60	µA
IDDA (excluding reference current)		0.3	1.0	mA
IDDD (static)		5	50	µA

Absolute limits:

- VDDA: 0 to 3.6 V. VDDD: 0 to 1.95 V.
- References and VOUT: between VSS and VDDA, with VREFL < VREFH.
- Digital inputs: between VSS and VDDD.
- VOUT must not be driven or shorted.
- IBIAS: ≤ 2 µA.

The Sky130 device and pad ratings take precedence over these limits.

5. Simulation to date

An LTSpice model of the architecture (64-resistor string, the transmission-gate tree, the level shifters, and enable selector) was run. These use generic Level-1 CMOS models, not Sky130 models. The buffer is the MCP6001 teaching macromodel.

Result	Value	What it shows
All 64 codes, monotonic	Pass; minimum step 42.187 mV	The tree and level-shifter logic select correctly
Worst absolute error	0.107 LSB (4.5 mV)	Almost all of this is the macromodel's built-in 4.5 mV offset. It sets the scale of the buffer-offset budget
DNL / INL	≈ 0	Expected with ideal, identical resistors. Silicon DNL/INL will depend on how well the Sky130 poly resistors match.
Worst settling (tested transitions, including 31↔32 and 0↔63)	6.2 µs	With a 1 MHz amplifier, the 20 µs target leaves margin
100 kΩ load placed directly on the tap	267 mV error (≈ 6 LSB)	The buffer is necessary
1 kΩ output load	Code 63 clips to 2.54 V	Shows the output-drive limit. A 1 kΩ load is outside the specified load range

These results validate the architecture and control logic only. The next step is to port the design to Xschem and ngspice with Sky130 devices, and to design the transistor-level buffer. The buffer is the main design task: it must accept inputs from 0.3 to 3.0 V on a 3.135 V supply and stay stable into 20 pF.

6. Basis for the chosen values

Several values are starting points, to be confirmed or revised against Sky130 simulation before the schematic review:

- **0.3 V and 3.0 V references:** chosen to keep the buffer 0.3 V away from each rail at nominal supply. At VDDA = 3.135 V, the top code leaves only about 0.18 V of headroom for the buffer output. If the buffer cannot meet spec there, we will lower VREFH (for example to 2.7–2.8 V), or specify the top codes at nominal supply only.

- **1 kΩ unit resistor (64 kΩ total):** sets a 42 µA reference current. This is a starting balance between reference current, resistor area, and settling. A higher resistance (for example 256 kΩ, about 10 µA) remains an option.
- **±0.5 LSB accuracy and 20 µs settling:** standard targets for a 6-bit DAC, set with margin above the LTspice results. They are not derived from a specific application requirement.
- **100 kΩ ∥ 20 pF load:** represents a typical on-chip or bench input. 20 pF is our estimate of the pad, package, and fixture capacitance.
- **Supply tolerance (±5%) and temperature range:** assumed until the harness documentation is published.
- **IDDA estimate and 1 µA bias:** the IDDA figure comes from first hand calculations for a two-stage op-amp. The 1 µA bias follows the harness guidance.
- **Transistor sizes in the LTspice model:** illustrative only, not the final Sky130 sizing.

7. Test plan outline

Test	**Method**	**Pass criterion**
Function	Step through all 64 codes over SPI; toggle EN	Correct level for each code; EN low gives VREFL
Transfer, DNL, INL	Precision DMM on VOUT, VREFH, and VREFL; sweep codes up and down; allow ≥ 1 ms per code	Absolute error, DNL, and INL within spec; monotonic
Buffer vs. string	Measure VTAP (≥ 10 MΩ input) and VOUT at each code	Separates resistor mismatch from buffer offset
Settling and glitches	Oscilloscope on VOUT; code steps 0↔63 and 31↔32, triggered on the output step	Settling ≤ 20 µs; glitch amplitude reported
Load	Repeat transfer at 100 kΩ and 1 MΩ, with 20 pF total	Within spec
Supply and reference current	SMU on VDDA, VDDD, VREFH, and VREFL; sweep IBIAS over its range	Within spec; stable across the IBIAS range
Temperature and supply	Repeat transfer and current tests at the supply and	Report against spec

Test	Method	Pass criterion
	temperature extremes available	

The references are driven by a source/measure unit. A plain bench supply cannot sink the VREFL current.